Mohammad Affaiq Aini

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Education

Universiti Teknologi MARA
Msc, Master's in Science Information Technology. **2026**
Thesis: Image Detection for Monitoring Soft Shell Crab Using Internet of Things
Publication(s): Image Detection Using YOLOv7 Algorithm to Detect the Moulted Crabs (e-proceedings book ICoBTA-IR4.0), Image Enhancement for Detection of Underwater Moulted Crabs in Greenhouse Soft-Shell Crab Farming Using Deep Learning, https://doi.org/10.1016/j.atech.2025.101223 (Q1 Journal - Smart Agricultural Technology).

Bachelor's Degree (Hons.) Chemistry with Management. **2022**

Diploma in Industrial Chemistry **2019**

Experiences

Arc Technologies Sdn Bhd Kuching, Sarawak
System Engineer January 2025 – Present

- Design and document system architecture for IoT-based solutions and operational systems.
- Prepare Product Requirement Documents (PRD) and technical documentation to support system implementation and governance.
- Deploy and manage virtual machine environments for system testing, simulation, and operational support.
- Install and configure Operational Technology (OT) simulators for industrial system testing and validation.
- Conduct site visits to support system deployment, troubleshooting, and technical assessments.
- Coordinate with clients and stakeholders to gather requirements and ensure system implementation meets operational needs.
- Support system integration and monitor system performance to ensure reliability and compliance with project requirements.

Karuna Sarawak Sdn Bhd Kuching, Sarawak
System Programmer June 2024 – December 2024

- Developed and maintained web-based systems using Laravel (PHP).
- Designed database structures and Entity Relationship Diagrams (ERD) to support system development and data management.
- Customized and implemented Odoo ERP modules to improve business workflow automation.
- Developed backend scripts using Python for system automation and data processing.
- Conducted system testing, debugging, and performance optimization to ensure application stability.

UiTM Sarawak Kota Samarahan, Sarawak
Research Assistant October 2022 – March 2024

- Specialize in IoT applications for object detection and image recognition in soft-shell crab and crayfish farming.
- Optimizing detection methods, creating image recognition models, collaborating with IoT technologies for seamless integration, and conducting data analysis.
- Discussing with the farm manager to provide needed assistance for the project.
- Became an Exhibitor for IoT Monitoring System Project at WCiT 2023
- Became a Presenter for IoT Monitoring System Project at iCOBTA 2023

Kompande Group Kota Samarahan, Sarawak
IT Support/Technician (Part time during semester break) 2021 – 2022

- Developed crucial skills for meticulous fiber-optic network installation and maintenance.
- Troubleshooting and diagnosing issues using testing equipment.
- Interacting with clients whenever they need services.
- Been outstation outside Kuching (Miri, Sibu, Lundu, Bau, Sematan) for technical maintenance.

Certifications

- AWS Certified Cloud Practitioner – Amazon Web Services
- Drone Pilot Certification (Aerial Photography)

Skills & Interests

Programming Language(s): Python, HTML, CSS, JavaScript, PHP, MySQL
Machine Learning: OpenCV, Tensorflow, Pytorch
Framework: Laravel, Flask, ReactJS, Typescript
Language: Bahasa Malaysia, English
Interests: Internet of Things (IoT) technologies, System design and technology integration, Data analysis and problem solving, Client engagement and technology consulting